

Formulario NGINX

¿Qué es NGINX?

NGINX es:

- Web Server
- Reverse Proxy
- Load Balancer
- API Gateway
- Proxy para contenedores

Muy usado con:

- Docker
- Kubernetes
- NodeJS
- FastAPI
- Flask

Arquitectura

```
Cliente
|
V
NGINX
|
+--> Backend 1
|
+--> Backend 2
```

Instalación

```
sudo apt update
```

```
sudo apt install nginx
```

Verificar:

```
nginx -v
```

Comandos importantes

Comando	Descripción
nginx -t	Validar configuración
systemctl start nginx	Iniciar
systemctl stop nginx	Detener
systemctl restart nginx	Reiniciar
systemctl reload nginx	Recargar config
journalctl -u nginx	Logs

Estructura Linux típica

```
/etc/nginx/  
|  
|-- nginx.conf  
|-- sites-available/  
|-- sites-enabled/  
|-- conf.d/
```

nginx.conf básico

```
events {}  
  
http {  
    server {  
        listen 80;  
        server_name localhost;  
        location / {  
            root /var/www/html;  
            index index.html;  
        }  
    }  
}
```

Bloques importantes

Bloque	Función
events	Conexiones/eventos
http	Config HTTP
server	Virtual Host
location	Rutas
upstream	Load balancing

Servir archivos estáticos

```
server {  
    listen 80;  
    server_name localhost;  
    location / {  
        root /var/www/html;  
        index index.html;  
    }  
}
```

Reverse Proxy

```
location /api/ {  
    proxy_pass http://localhost:5000/;  
}
```

NGINX recibe requests y las reenvía.

Headers importantes

```
location / {  
    proxy_pass http://backend;  
    proxy_set_header Host $host;  
    proxy_set_header X-Real-IP $remote_addr;  
    proxy_set_header X-Forwarded-For  
    $proxy_add_x_forwarded_for;  
}
```

Load Balancer

```
upstream backend {  
    server localhost:5000;  
    server localhost:5001;  
}  
  
server {  
    location / {  
        proxy_pass http://backend;  
    }  
}
```

Docker + NGINX

```
location / {  
    proxy_pass http://web_container:8000;  
}
```

NGINX puede comunicarse usando nombres de contenedores.

HTTPS

```
server {

    listen 443 ssl;

    ssl_certificate
    /etc/ssl/cert.pem;

    ssl_certificate_key
    /etc/ssl/key.pem;

}
```

Redirección HTTP -> HTTPS

```
server {

    listen 80;

    return 301 https://$host$request_uri;

}
```

Logs

```
/var/log/nginx/access.log
/var/log/nginx/error.log
```

Validar configuración

```
sudo nginx -t
```

MUY importante antes de reiniciar.

Recargar configuración

```
sudo systemctl reload nginx
```

No mata conexiones activas.

Errores comunes

Error	Posible causa
502 Bad Gateway	Backend muerto
bind() failed	Puerto ocupado
Permission denied	Permisos Linux
404	Ruta incorrecta

Proxy para WebSockets

```
location /socket {

    proxy_pass http://backend;

    proxy_http_version 1.1;
```

```
proxy_set_header Upgrade $http_upgrade;

proxy_set_header Connection "upgrade";

}
```

Rate Limiting

```
limit_req_zone
$binary_remote_addr
zone=api:10m
rate=5r/s;
```

Workers

```
worker_processes auto;
```

NGINX crea workers según CPUs.

Flujo típico

```
Cliente
|
V
NGINX
|
+--> React Frontend
|
+--> FastAPI Backend
|
+--> PostgreSQL
```